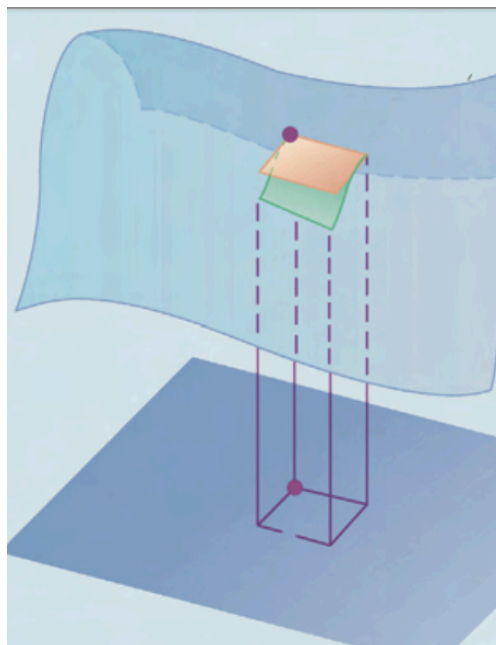


MATH 280 MATLAB Project #5 - Vector Field and Surface Analysis Studio



When the project is completed, rename the end of the file to (Completed Implementation).



PROJECT OUTLINE

Core Objective

Create a MATLAB application that allows users to visualize and analyze vector fields, line integrals, surface integrals, conservative fields, and Green's Theorem through an interactive graphical user interface. The application provides engineering-focused tools for computing work, circulation, flux, and surface area while displaying dynamic vector field visualizations.

Concepts Covered

Chapter 16 Integration in Vector Fields

16.1 Line Integrals

16.2 Vector Fields

16.3 Conservative Fields

16.4 Green's Theorem

16.5 Surface Area

Additional topics:

- Work
- Circulation
- Flux
- Potential Functions
- Path Independence

Learning Objectives

By completing this project you will practice:

- vector field analysis
- line integrals
- surface integrals
- numerical integration

- **engineering visualization**
- **physical modeling**
- **MATLAB App Designer**
- **interactive scientific computing**

Required Features

1. Vector Field Explorer in MATLAB

User enters a function of two/three variables

MATLAB displays:

- vector arrows
- streamlines
- field magnitude coloring
- field direction

This feature supports predefined fields such as radial, rotational, gravitational, and electric field models.

2. Line Integral Calculator

User enters:

- vector field
- parametric curve

MATLAB computes:

- line integral
- accumulated quantity along the path

And displays:

- highlighted path
- moving integration point
- numerical result

3. Work Calculator in MATLAB

User selects force field and path

MATLAB computes:

- total work performed
- positive and negative work regions

And displays:

- force vectors
- trajectory
- cumulative work graph

4. Circulation and Flux Analyzer in MATLAB

MATLAB computes:

- circulation around closed curves
- outward flux across boundaries

And displays:

- flow direction
- circulation animation
- flux vectors

This feature provides side-by-side comparisons of circulation and flux for the same field.

5. Conservative Field Explorer

MATLAB calculates and determines:

- whether a field is conservative
- potential function (if one exists)
- path independence

And displays:

- gradient field
- potential surface
- comparison of multiple paths between identical endpoints

6. Green's Theorem Visualizer in MATLAB

User enters

- closed curve
- vector field

MATLAB computes

- line integral
- corresponding double integral

And displays:

- enclosed region
- boundary traversal
- numerical comparison verifying Green's Theorem

7. Surface Area Calculator in MATLAB

User enters the surface equation

MATLAB computes:

- surface area

- surface mesh

And displays:

- shaded surface
- tangent plane approximation
- surface normals

8. Surface Integral Analyzer in MATLAB

MATLAB computes

- scalar surface integrals
- vector flux integrals

And displays:

- integration surface
- normal vectors
- colored contribution map
- accumulated integral

9. Field Animation Studio in MATLAB

Animate in MATLAB:

- moving particles
- streamline tracing
- force vectors
- changing field intensity

Users can adjust:

- animation speed
- particle count
- trail length

10. Engineering/Physics Application Library

Include built-in demonstrations such as

- gravitational fields
- electric fields (Physics 2 E&M)
- magnetic field approximations (Physics 2 E&M)
- fluid flow around obstacles
- heat flow
- wind velocity fields
- force acting on a particle
- pressure distribution

This is similar to previous projects.

Users can modify parameters and compare multiple scenarios with this feature.

Necessary MATLAB Files For This Project/File Structure

MVC_Project5/

main.mlapp

main.m

vectorFieldGenerator.m

lineIntegralCalculator.m

workCalculator.m

circulationCalculator.m

fluxCalculator.m

potentialFunctionAnalyzer.m

greensTheoremVerifier.m

surfaceAreaCalculator.m

surfaceIntegralCalculator.m

fieldAnimator.m

plotUtilities.m

README.md

Useful MATLAB Commands For This Project

quiver()
quiver3()
streamline()
streamslice()
coneplot()
surf()
mesh()
fsurf()
fplot3()
integral()
integral2()
gradient()
curl()
divergence()
scatter3()
plot3()
hold on
axis equal
view()
uifigure()
uiaxes()
uibutton()
uidropdown()
uislider()
uitable()
uicontrol()
camlight()

lighting()

Additional MATLAB Pseudocode

Vector Field Generation

Receive vector field

Generate grid

Evaluate field

Plot arrows

Apply magnitude coloring

Line Integral

Receive path

Evaluate field

Compute tangent vectors

Evaluate integral

Display result

Work Calculation

Receive force field

Receive path

Evaluate line integral

Display total work

Conservative Field Test

Receive vector field

Evaluate conditions

Determine if conservative

Compute potential function

Verify path independence

Green's Theorem Verification

Receive closed curve

Compute line integral

Compute double integral

Compare values

Display verification

Surface Integral

Receive surface

Generate mesh

Compute normals

Evaluate integral

Visualize contributions

GUI FUNCTIONALITY

This should be the most sophisticated GUI out of all 5 MVC projects:

- Interactive vector field editor
- Drop-down menu of predefined physical fields
- Streamline visualization toggle
- Particle animation controls
- Path drawing tool for custom curves
- Surface selector for common geometries
- Checkboxes to display:
 - Vector arrows
 - Streamlines
 - Surface normals
 - Flux vectors
 - Potential contours
 - Integration paths
- Split-screen mode for comparing two vector fields
- Export plots, animations, and numerical results

MAIN MENU

Vector Field Analysis Studio

- 1. Create Vector Field**
- 2. Line Integrals**
- 3. Work Calculator**
- 4. Circulation and Flux**
- 5. Conservative Field Analysis**
- 6. Green's Theorem**
- 7. Surface Area**
- 8. Surface Integrals**
- 9. Engineering Applications**
- 10. Export Results**
- 11. Exit**

GUI LAYOUT

Primary Interface

Left Panel

Input fields

- **Vector field**

- Parametric curve
- Surface equation
- Integration limits
- Path selection
- Surface selection

Buttons

- Plot
- Analyze
- Animate
- Reset

Center

Interactive visualization

- vector field
- streamlines
- surfaces
- curves
- field animations

Right Panel

Displays

- line integral
- work
- circulation
- flux
- potential function
- surface area
- surface integral

- Bottom

Buttons

- Rotate
- Zoom
- Export
- Save Plot
- Animation Controls

multivariable calculus

$$\iiint f(x,y,z) \partial f / \partial y \partial f / \partial x 7i + 4j + 3z$$